MIL-B-45852A
29 September 1965
SUPERSEDING
MIL-B-45852
14 July 1960
MIL-B-45853
14 July 1960

MILITARY SPECIFICATION

BORESCOPES M1 AND M2

This specification is mandatory for use by the Departments of the Army and the Navy of the Department of Defense

1. SCOPE

1.1 This specification covers two sizes of borescopes for inspecting bores 20MM and over.

1.2 Classification. Borescopes covered by this specification shall be of the following types:

a. M1 - for bores of 20-57MM
b. M2 - for bores over 57MM

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-G-174	-	Glass, Optical.
MIL-C-675	-	Coating of Glass Optical Elements, Anti-Reflection.
MIL-A-3920	-	Adhesive Optical, Thermosetting.
MIL-G-10944	-	Gages Dimensional Controls.
MIL-O-13830	-	Optical Components for Fire Control Instruments, General Specification Governing the Manufacture, Assembly and Inspection of.
MIL-P-14061	-	Packaging of Borescopes M1 and M2.

STANDARDS

Military

MIL-STD-10	-	Surface Roughness, Waviness and Lay.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-130	-	Identification Marking of U. S. Military Property.

FSC 4933

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DRAWINGS

U. S. Army Weapons Command
EPL 8770505 - Engineering Parts List for Borescope M1.
EPL 8770661 - Engineering Parts List for Borescope M2.
IEL 8770505 - Index to List of Inspection Equipment for Borescope M1.
IEL 8770661 - Index to List of Inspection Equipment for Borescope M2.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards
Handbook H-28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C., 20325)

3. REQUIREMENTS

3.1 Initial production samples. On each contract, samples of the borescope and parts representing initial production shall be taken by the Quality Assurance Representative and shall be forwarded to the Commodity Arsenal. Samples shall be representative of material which has been inspected and approved by the Q.A. Rep.

3.2 Materials and construction. Borescopes and parts thereof shall conform to this specification, the applicable drawings and specifications referenced in Engineering Parts Lists EPL 8770505 and EPL 8770661 and specifications MIL-O-13830, MIL-G-174, MIL-C-675, and MIL-A-3920.

3.3 Design. Borescopes and parts shall conform to the design specified on the applicable drawings. If it should appear that dimensions in addition to those specified on the applicable drawing are required, the supplier shall apply to the Q.A. Rep. for clarification. In no case shall drawings be scaled to obtain dimensions.

3.3.1 Optical system

3.3.1.1 Eyepiece focus. For an axial object at infinity it shall be possible to focus the eyepiece at -4 diopters. For an axial object located 3/4 inch for the M1 and 2 inches for the M2 from the objective end of the objective tube it shall be possible to set the eyepiece focus at +4 diopters. This range of adjustment shall be obtainable with any or all extension tubes added to the objective tube for the M2. For the M1, if the rack lies between 0.573 and 1.077 inches from the end of the sleeve when the eyepiece is focused on infinity, the above will be obtainable.

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3.3.1.2 Image size. The image of a 5.462-inch diameter circle for the M1 and a 6-inch diameter circle for the M2 at 10 inches from the objective end of the objective tube shall be between 0.440 inch and 0.660 inch in diameter at the field stop of the eyepiece.

3.3.1.3 Image location. The image of an axial object at infinity (more than 10 feet for the M1 and 15 feet for the M2 from the objective end of the objective tube) shall be between 0.400 inch and 1.000 inch beyond the male end of the eyepiece cup for the M1, and between 3.850 and 4.060 inches beyond the male end of the M2 objective tube assembly. Further, for each individual extension tube of the M2, the image of an object placed 8.360 inches into the female end of the extension tube shall lie between 3.850 and 4.060 inches from the male end of the extension tube.

3.3.1.4 Centering.

3.3.1.4.1 M-1. For the M1, cutoff of the field in the eyepiece by any apertures in the instrument shall be such that a 9.84-inch diameter circle placed at a distance of 10 inches from the end of the objective tube and at any radial position of the borescope, shall be fully visable.

3.3.1.4.2 M-2. For the M-2, cutoff of the field in the eyepiece by any apertures in the instrument shall be such that an 11.18-inch diameter circle placed at a distance of 10 inches from the end of the objective tube with any combination of extension tubes and at any radial position of the borescope, shall be fully visible.

3.3.1.5 Cleanliness. No dirt shall be visible in the focal plane throughout the range of eyepiece movement. With the eyepiece removed, loose dirt visible on the erector lenses shall not exceed 1.0 square millimeter in total area.

3.3.1.6 Resolution. As examined with a 4-power auxiliary telescope, the axial resolution shall not be less than 3.5 minutes for the M1 and 2 minutes with any combination of extension tubes for the M2 borescope.

3.3.2 Electrical. Insulation resistance as measured with a megohm bridge or megger-type insulation tester capable of generating 500 volts d.c. shall be at least the following:

a. M1 - 50 megohms between primary and case of transformer; 50 megohms between primary and secondary of the transformer; 50 megohms between central "pin" and shell of connector on lead cord, with cord switch in the "ON" position; 20 megohms between center pin and shell of collector housing connector with the bulbs removed from the head.

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b. M2 - 50 megohms between primary and case of transformer; 50 megohms between primary and secondary of the transformer; 50 megohms between prongs of the twist lock plug with the cord switch in the "ON" position; 20 megohms between current carrying parts on the adaptor assembly, the objective tube, all extension tubes, and for all illuminating heads with lamps removed.

3.3.3 Functional. The assembled borescope shall be given functional tests in bores from 20MM to 57MM for the M1 and 57MM and over for the M2. The quality of the image shall be such as to permit a critical inspection of the bores for defects such as small scratches, cracks, flaws, etc. These defects shall be clearly visible in the mirror and circumferential images.

3.3.4 Interchangeability. Assemblies and subassemblies shall be interchangeable with like units. Such parts thereof as are replaceable without damage other than to cotter pins, locking wire and the like shall be individually interchangeable. The following subassemblies of the borescope shall be interchangeable:

M1

a. Illuminating Assembly

(1) Head Assembly - 20MM - 37MM
(2) Head Assembly - 37MM - 57MM

b. Eyepiece Assembly

c. Main Tube Assembly

d. Transformer Assembly

M2

a. Illuminating Assembly

(1) Head Assembly - 57MM - 105MM
(2) Head Assembly - 105MM and over
(3) Prism Head Assembly - 57MM and over

b. Objective Tube Assembly

c. Extension Tube Assembly

d. Adapter Tube Assembly

e. Eyepiece Assembly

f. Transformer Assembly

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3.3.5 Marking. Marking shall be in accordance with MIL-STD-130. Marking shall be located as specified by the drawings. When location is not specified marking shall be placed, if possible, on a non-bearing surface in an area easily seen after assembly.

3.3.6 Workmanship.

3.3.6.1 Functioning surfaces. Contact surfaces shall be free from steps, projections, or other irregularities that may affect the functioning or fit of parts. Burrs and fins shall be removed to avoid hazards in handling.

3.3.6.2 Exposed surfaces. Surfaces not contacting other parts or units shall be true to specified contours and free from visible irregularities or defects that may adversely affect strength or serviceability or seriously detract from good appearance.

3.3.6.3 Surface roughness, waviness, and lay. The roughness, waviness, and lay of surfaces shall be as specified on the drawings. The standards and conditions of MIL-STD-10 shall delineate the requirements.

3.3.6.4 Protection. All surfaces shall be effectively protected against rust, corrosion, and physical injury during manufacture and assembly and until delivered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own inspection facility or any other inspection facility acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions as used in this specification are in accordance with MIL-STD-109.

4.3 Initial production samples. Inspection of the initial production samples shall be performed at the responsible Commodity Arsenal. Initial production samples shall be subjected to the inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract. Acceptance of regular production shall be withheld until the initial production samples have been approved.

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4.4 Inspection provisions.

4.4.1 One hundred percent inspection. Each borescope shall be examined for finish, workmanship, and function; and completeness of manufacture and assembly.

4.4.2 Lot formation. The formation, size and presentation of lots of the borescopes, parts, subassemblies and assemblies for inspection shall be as determined or approved by the Q.A. Rep. and subject to the provisions of MIL-STD-105.

4.4.3 Parts and subassemblies. Inspection of parts and subassemblies for use in assembly of the borescopes or to be furnished as repair or replacement parts shall be subjected to standard sampling inspection conducted in accordance with MIL-STD-105. All units of each lot shall be inspected for those requirements for which 100 percent inspection has been prescribed and defective units shall be removed. Sample selection and sample sizes for dimensional and non-dimensional characteristics, except physical requirements, shall be based on inspection level II of MIL-STD-105. For physical requirements, inspection level S-2 of MIL-STD-105 shall be used.

4.5 Tests.

4.5.1 Materials. Materials for use in the parts shall be inspected in accordance with the applicable material specification shown on the part drawing. Required tests may be conducted in the prime contractor's or sub-contractor's facilities. The test results shall be available to the Q.A. Rep. for review. Certified test reports, identifiable with the material may, at the option of the Government, be accepted in lieu of tests. When the identity or quality of the material is in doubt, and in the absence of valid and acceptable test data, the supplier shall conduct or have conducted such tests within the scope of the material specifications as are required by the Q.A. Rep. to determine the identity and quality.

4.5.2 Optical parts. Optical parts shall be inspected by Government approved optical test methods and equipment. In the absence of approved optical test methods and equipment, the following typical procedures shall be adhered to:

4.5.2.1 Finish and defect test. Finishes and defects of each lens and prism shall be checked for compliance with MIL-G-174 by viewing, using one of the following methods:

(a) Inspection method No. 1. The lens or prism to be inspected shall be viewed against a ground or opal glass surface illuminated from behind by a 40-watt bulb approximately 3 inches from the glass surface. Two or more opaque horizontal bars occupying approximately 1/2 of the area of the ground or opal glass surface shall be placed in front of and in contact with the glass surface.

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(b) Inspection method No. 2. Light from a 40-watt bulb shall be passed through a ground or opal glass and then through the lens or prism. Defects shall be observed by light scattered from the surface while viewing it against a dark background.

4.5.3 Definition test. The definition test shall be standard and shall be performed using one of the National Bureau of Standards Resolution Charts. Resolving power is a measure of the optical performance of a telescope. The resolving power of a telescope is the angular subtense (in minutes of arc) of a series of parallel bars which can just be resolved.

4.5.4 Definition tests for optical components or partial systems. Optical components or partial systems procured as such and not as complete systems shall be tested for performance and definition by comparison with acceptable samples. The Government may furnish the samples or the supplier may be required to select from his manufacture two samples for each type of element which samples shall be sent to a responsible technical activity for approval. The Government shall retain one sample for each type of element.

4.5.5 Protective coatings. Tests of the protective coating on parts for which protective coating is specified on the drawings shall be conducted in accordance with the applicable drawings and coating specification. The tests shall be repeated when visual examination indicates lack of conformance with the appearance requirements of the applicable coating specification. A minimum of one sample specimen for each applicable test shall be selected at random from the lot being inspected. The lot shall be subject to rejection if the specimens fail in any of the tests.

4.5.6 Electrical tests. Insulation resistance shall be measured with a megohm bridge, megger-type tester, or similar equipment. Continuity of applicable circuits shall be assured using an ohmmeter, a light bulb or similar indicator.

4.5.7 Functioning tests. Each borescope and major subassembly shall be tested for functioning, using the test methods specified. Failure of the borescope or subassembly to meet the functional requirements shall be analyzed and the borescope or subassembly reconditioned, where indicated, prior to any retest.

4.6 Examinations.

4.6.1 Diameter of exit pupil. The diameter of the exit pupil shall be measured by means of an auxiliary eyepiece mounted in a draw tube with a reticle or a suitably graduated scale. The reticle shall be in the focal plane of the auxiliary eyepiece. In operation, the auxiliary eyepiece is placed in front of the borescope eyepiece and moved in or out

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until the exit pupil is in sharp focus on the reticle scale. The diameter of the pupil is then read directly from the scale. Eye distance of exit pupil shall be measured when the exit pupil is brought into sharp focus on the auxiliary eyepiece reticle in measuring the diameter of the exit pupil. The eye distance is the distance from the auxiliary reticle to the eyelens of the borescope.

4.6.2 Diameter of entrance pupil. The entrance pupil shall be measured by means of a traveling microscope. The borescope to be measured shall be placed at a convenient distance with the axis perpendicular to the line of travel of the measuring microscope. The entrance pupil can then be measured by bringing the crosshair of the traveling microscope in coincidence with each step of the pupil and noting the distance traveled.

4.6.3 Magnification. A satisfactory determination of magnification shall be made by dividing the diameter of the entrance pupil by that of the exit pupil.

4.6.4 The field of view. The field of view shall be measured by viewing a properly calibrated distant target scale across a diametral line of the field of view.

4.6.5 Interchangeability. The Q.A. Rep. shall subject a sample of three borescopes, randomly selected from each lot submitted for acceptance to an interchangeability test. The three borescopes, each complete and individually acceptable, shall be disassembled into subassemblies. The subassemblies shall be randomly interchanged and reassembled. The resulting three complete borescopes shall be tested for functioning. Failure in the interchangeability test shall be cause for reinspection or rejection of the represented lot.

4.6.6 Surface roughness. Surface roughness, interpreted in accordance with MIL-STD-10, shall be visually inspected by direct comparison with applicable surface roughness specimens. In case of dispute, surface finish measuring equipment acceptable to the Q.A. Rep. shall be used.

4.6.7 Threads. Unless otherwise specified, the dimensions of all screw-threads shall conform to the limits shown in the National Bureau of Standards Handbook H-28. Threads shall be checked with thread gages appropriate to the type, form, size, and class of threads specified.

4.6.8 Protective coatings. Appearance of protective coatings shall be visually examined for conformance with the workmanship requirements of the applicable coating specification designated on the drawing. The coating shall be evenly deposited, uniform in color, free from pits, bare spots, or corrosion and shall not exhibit a mottled appearance.

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4.6.9 Cleanliness. Each borescope shall be examined through the objective and eyepiece ends with the unaided eye. Inspection for moisture shall be made by the technique of shadowing. Inspection for dust particles shall be made by viewing a uniformly illuminated field having a brightness of approximately 300 apparent foot-candles.

4.6.10 Marking. Items shall be visually examined to determine compliance with marking requirements specified on the applicable drawings. Markings shall be complete, correct, and legible.

4.6.11 Workmanship. Items shall be inspected visually to determine compliance with the requirements of this specification. Missing operations and the presence of nicks, scratches, burrs, toolmarks, deformations, and knife edges shall be considered as evidence of poor workmanship.

4.7 Inspection equipment.

4.7.1 Ordnance inspection equipment. Ordnance drawings or stock numbers of the final inspection equipment, other than standard measuring instruments, needed to determine compliance of a part, subassembly or assembly with the requirements of the applicable drawings and specifications will be obtained from the applicable gage list given below:

Index to List of Inspection Gages and Parts Gaged on Borescope M1	IEL 8770505
Index to List of Inspection Gages and Parts Gaged on Borescope M2	IEL 8770661

When specified in the contract or by the contracting officer, special inspection equipment thus referenced will be available for joint usage by the supplier and the Q.A. Rep. for performing examinations and tests.

4.7.2 Supplier's final inspection equipment. Unless otherwise specified in the contract or by the contracting officer, the supplier shall provide and maintain gages in conformance with MIL-G-10944, and other measuring and testing devices considered necessary for supplier's inspection of parts, subassemblies and assemblies to substantiate conformance to contractual requirements. Such equipment will be subject to inspection by the Government and will be made available to the Q.A. Rep., when requested for the purpose of product inspection.

In inspecting the supplier's inspection equipment, the Q.A. Rep. shall determine that the supplier has available, and uses correctly, gaging, measuring, and testing equipment of required accuracy and precision, and that the instruments are of proper type and range to make measurements of the required accuracy. The supplier shall have available a set of master gages, standards, and appropriate

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instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the supplier and made available for review by the Q.A. Rep. The calibration of gages, standards and instruments shall be subject to periodic check by the Q.A. Rep. or when deemed necessary.

5. PREPARATION FOR DELIVERY

5.1 Procedure. The preparation for delivery shall be in accordance with MIL-P-14061.

6. NOTES

6.1 Intended use. The borescopes are intended for visually inspecting items such as cannon bores, propellant perforations of rocket motors, and similar items.

The M1 borescope is used to inspect bores from 20 to 57MM in diameter. It is a single tube instrument and can be used to inspect 20MM bores of lengths up to approximately 66 inches. Bores larger than 1.425 inches in diameter can be inspected up to lengths of approximately 76 inches. Support rings are provided for centering the instrument.

The M2 borescope is used to inspect bores 57MM and larger in diameter. Three extension tubes containing appropriate optical systems are normally provided to permit inspection of bores of lengths up to approximately 26 feet. This length can be increased by the use of additional extension tubes. Support rings are provided for centering the instrument.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) List of repair parts required.

(c) Levels of packaging and packing protection required.

(d) Availability of inspection equipment from the U. S. Army Weapons Command.

(e) Extent of contractor's responsibility for Government-furnished and for contractor-acquired final inspection equipment.

(f) Number of initial production samples required.

(g) That inspection system requirements shall conform to MIL-I-45208.

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6.3 International standardization agreement. The items covered by this specification are the subject of International Standardization Agreement, ABCA-ARMY-STD-148. When any amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the International agreement concerned.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army-WC

Preparing activity:

Army--WC

Project No. 4933-0014

Reviewer activity:

Army--WC

User activities:

Army--MI
Navy--MC

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF 'YES IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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